Classical Mechanics

Midterm Exam

November 21st, 2023

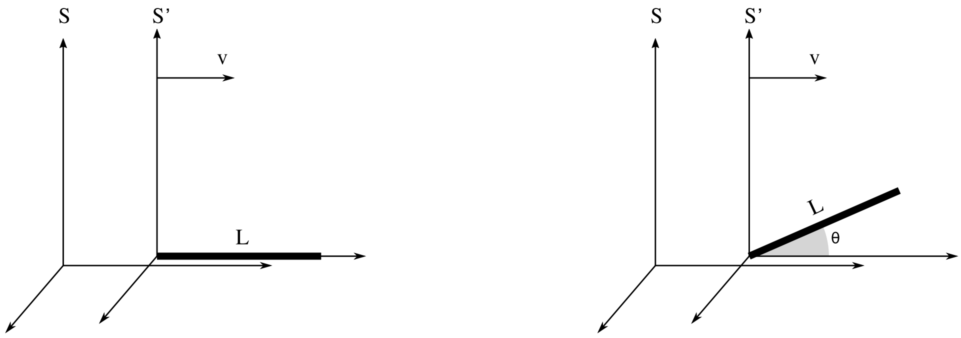
Question 1 (1 point)

Consider some general coordinates $\vec{x}$, and a new coordinate system $\vec{q}$, which is related to $\vec{x}$ by the change of coordinates $\vec{q} = \vec{q}(\vec{x}, t)$. Prove that

$$\frac{\partial \dot{q}_i}{\partial \dot{x}_j} = \frac{\partial q_i}{\partial x_j} \,.$$

Exercise 1 (3 points)

A stick of proper length L moves past you with relativistic velocity $\vec{v}$.



Situation A　　　Situation B

In **Situation A**, the motion of the stick is parallel to the stick itself, as shown in the left panel of the figure. In this situation, there is a time interval between the front end of the stick coinciding with you and the back end coinciding with you.

a) What is this time interval in your frame S?

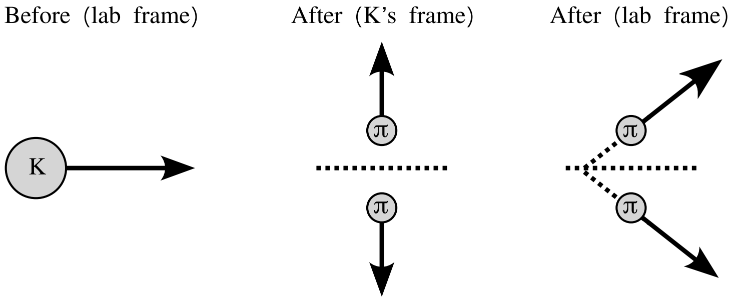
b) What is this time interval in the stick's frame S'?

The situation is slightly modified in **Situation B**. Now, the stick moves past you inclined at an angle θ in S', as shown in the right panel of the figure.

c) In this new situation, what is the length of the stick as measured in your frame S?

Exercise 2 (3 points)

A kaon K with mass M and an initial velocity v (as seen from the lab frame) decays into two pions π, each of which with mass $m < M/2$. Suppose that, in the rest frame of the kaon, the pions are emitted in opposite directions and perpendicular to the initial propagation direction of the kaon, as shown in the figure below.

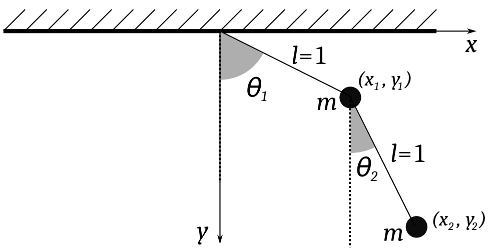


a) Find the energy-momentum 4-vectors for the pions in the kaon rest frame.

b) Find the energy-momentum 4-vectors in the lab frame, and the angle of the trajectory of the pions with respect to the original propagation direction of the kaon.

Exercise 3 (3 points)

Consider the double pendulum shown in the following figure. Assume that both masses are equal to m and that the length of both pendulums is $l = 1$.



a) Write the Lagrangian of the system as a function of the angles θ_1 and θ_2, as defined in the figure. **Hint**: You may want to use the identity $\cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 = \cos(\theta_1 - \theta_2)$ to simplify your expressions and calculations.

b) From the Lagrangian, obtain the equations of motion for θ_1 and θ_2.

Q.1

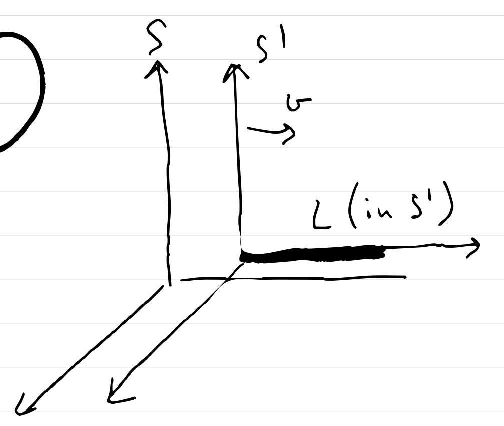
$$\frac{\partial \dot{q}_i}{\partial \dot{x}_j} = \frac{\partial}{\partial \dot{x}_j} \frac{d q_i}{dt} = \frac{\partial}{\partial \dot{x}_j} \left[\sum_k \frac{\partial q_i}{\partial x_k} \dot{x}_k + \frac{\partial q_i}{\partial t} \right] =$$

$$= \sum_k \frac{\partial}{\partial x_k} \underbrace{\left(\cancel{\frac{\partial q_i}{\partial \dot{x}_j}} \right)}_{0} \dot{x}_k + \sum_k \frac{\partial q_i}{\partial x_k} \underbrace{\frac{\partial \dot{x}_k}{\partial \dot{x}_j}}_{\delta_{jk}} + \frac{\partial}{\partial t} \underbrace{\cancel{\frac{\partial q_i}{\partial \dot{x}_j}}}_{0}$$

because q_i only depends on $\vec{x}$, but not $\dot{\vec{x}}$!

$$= \frac{\partial q_i}{\partial x_j} \quad \square$$

E1



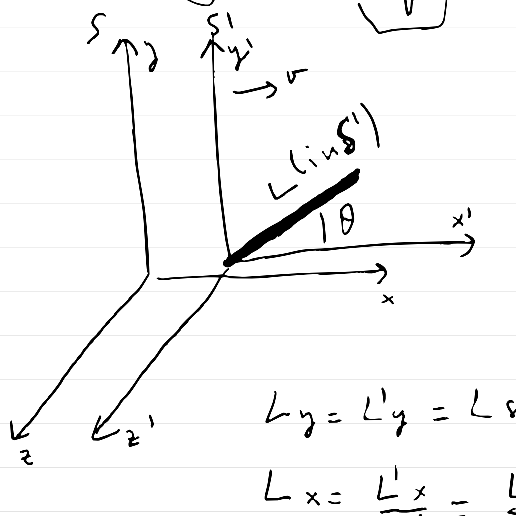
a) In S, the length of the stick is $\frac{L}{\gamma}$ (length contraction) and it is moving at v, so

$$\boxed{T = \frac{L/\gamma}{v} = L\frac{\sqrt{1-v^2}}{v} = L\sqrt{\frac{1}{v^2}-1}}$$

b) Since both events happen at the same point in S:

$$T' = \gamma T = \boxed{\frac{L}{v}}$$

time dilation



c) Length is contracted in x direction but not in y.

$$L'_x = L\cos\theta$$

$$L'_y = L\sin\theta$$

$$\left.\begin{array}{l} L_y = L'_y = L\sin\theta \\ L_x = \frac{L'_x}{\gamma} = \frac{L\cos\theta}{\gamma} \end{array}\right\} \ell = \sqrt{L_x^2 + L_y^2} = (\star)$$

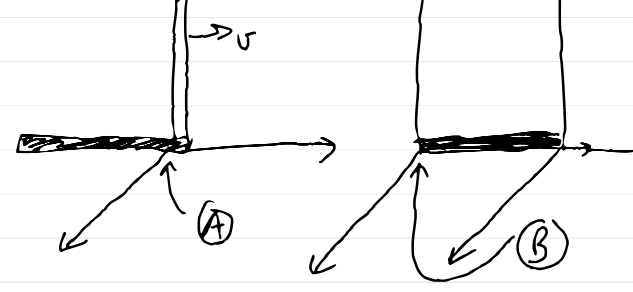
ℓ: length in S

$$(\star) = \sqrt{L^2\sin^2\theta + L^2\frac{\cos^2\theta}{\gamma^2}} = L\sqrt{\sin^2\theta + \cos^2\theta(1-v^2)} = \boxed{L\sqrt{1-v^2\cos^2\theta}}$$

Alternative methods for a).

(1) You may be inclined to use the general method of Lorentz transformations. Of course, that is fine — but you have to be careful!

First, if you proceed as we always did in class, you have to make sure that the origins coincide at (A)

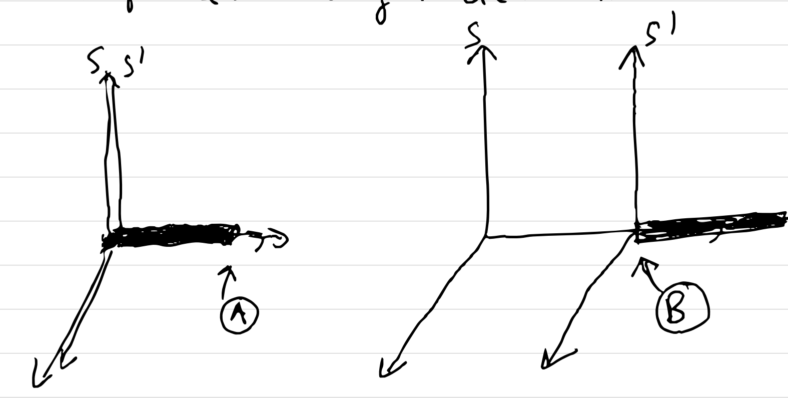


	S	S'
A	$x_A = t_A = 0$	$x'_A = t'_A = 0$
B		$x'_B = -L$, $t'_B = \frac{L}{v}$

Note that $x'_B = -L$ and that $t'_B = \frac{L}{v}$ (from the perspective of the stick, the observer is moving towards the left at v).

With this, we have $\boxed{T = t_B} = \gamma(t'_B + v x'_B) = \gamma\left(\frac{L}{v} - vL\right) = \frac{\gamma L}{v}(1-v^2) = \frac{\gamma L}{v\gamma^2} = \boxed{\frac{L}{v\gamma}}$

(2) If you do not want to make the origins coincide with the first event (A) then you can solve the problem using increments



	A	B
S	?	?
S'	$x'_A = L$, $t'_A = 0$	$x'_B = 0$; $t'_B = \frac{L}{v}$

$$\boxed{T = \Delta t_B} = \gamma(\Delta t'_B + v\Delta x'_B) = \gamma\left(\frac{L}{v} + v(-L)\right) = \frac{\gamma L}{v}(1-v^2) = \boxed{\frac{L}{\gamma v}}$$

(3) Finally, you may want to choose origins as in alternative (2), and not work with increments, but that is dangerous. Let's see why. Let's take x'_B and t'_B as in (2)

$$t_B = \gamma(t'_B + v x'_B) = \frac{\gamma L}{v}$$

and it seems that we are getting a different result!

However, note that, with this choice of origins:

$$t_A = \gamma(t'_A + v x'_A) = +\gamma v L \neq 0!!$$

Therefore

$$\boxed{T = t_B - t_A} = \frac{\gamma L}{v} - \gamma v L = \frac{\gamma L}{v}(1-v^2) = \boxed{\frac{L}{\gamma v}} \;\square.$$

Note that, with this choice of origins, both (A) and (B) happen at $x_A = x_B = \gamma L$!

E2 a) Before (K's frame) After (K's frame)

M $\bigcirc_K$ $\qquad$ $\pi_1 \bigcirc m$ (↑) $\quad$ $\pi_2 \bigcirc m$ (↓)

$P_K = (M, 0, 0, 0)$ $\qquad$ $P_1 = (E_1, 0, p_1, 0)$

$P_2 = (E_2, 0, p_2, 0)$

Conservation in y: $p_1 = -p_2 \equiv p \Rightarrow E_1 = E_2 \equiv E$

Conservation in t: $M = 2E \Rightarrow E = \frac{M}{2}$

Therefore: $P_1 = \left(\frac{M}{2}, 0, p, 0\right)$

$P_2 = \left(\frac{M}{2}, 0, -p, 0\right)$

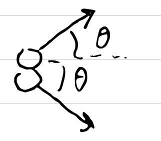
Additionally $-m^2 = P_1^2 = -\left(\frac{M}{2}\right)^2 + p^2 \Rightarrow p = \sqrt{\left(\frac{M}{2}\right)^2 - m^2}$

So finally:

$$\boxed{P_1 = \left(\frac{M}{2}, 0, \sqrt{\left(\frac{M}{2}\right)^2 - m^2}, 0\right) ; \quad P_2 = \left(\frac{M}{2}, 0, -\sqrt{\left(\frac{M}{2}\right)^2 - m^2}, 0\right)}$$

b) Since P_1 & P_2 are 4-vectors, we know they Lorentz-transform

→ Lab frame

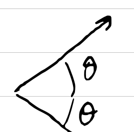


$$\left.\begin{aligned} P_1^{\prime 0} &= \gamma\left(P_1^0 + v P_1^1\right) = \gamma \frac{M}{2} \\ P_1^{\prime 1} &= \gamma\left(P_1^1 + v P_1^0\right) = \frac{\gamma v M}{2} \\ P_1^{\prime 2} &= P_1^2 = \sqrt{\left(\frac{M}{2}\right)^2 - m^2} \\ P_1^{\prime 3} &= 0 \end{aligned}\right\} \Rightarrow \boxed{P_1' = \left(\frac{\gamma M}{2}, \frac{\gamma v M}{2}, \sqrt{\left(\frac{M}{2}\right)^2 - m^2}, 0\right)}$$

$$\boxed{\theta} = \text{arctg} \frac{P_1^{\prime 2}}{P_1^{\prime 1}} = \text{arctg} \frac{\sqrt{\left(\frac{M}{2}\right)^2 - m^2}}{\frac{\gamma v M}{2}} = \text{arctg} \frac{\sqrt{M^2 - 4m^2}}{\gamma v M}$$

PAINFUL ALTERNATIVE

$P_K' = \left(\sqrt{M^2 + p^2}, p, 0, 0\right)$



$P_1' = \left(\frac{\sqrt{M^2+p^2}}{2}, p_1 \cos\theta, p_1 \sin\theta, 0\right)$

! But need to justify $\theta_1 = \theta_2 \equiv \theta$ and $p_1 = p_2$!

$P_2' = \left(\frac{\sqrt{M^2+p^2}}{2}, p_1 \cos\theta, -p_1 \sin\theta, 0\right)$

$$p^2 = M^2 \gamma^2 v^2 \Rightarrow \frac{\sqrt{M^2+p^2}}{2} = \frac{\sqrt{M^2(1+\gamma^2 v^2)}}{2} = \frac{M}{2}\sqrt{1 + \frac{v^2}{1-v^2}} =$$

$$= \frac{M}{2}\sqrt{\frac{1 - \cancel{v^2} + \cancel{v^2}}{1-v^2}} = \frac{M\gamma}{2} \Rightarrow$$

$$\Rightarrow \begin{cases} P_1' = \left(\frac{M\gamma}{2}, p_1 \cos\theta, p_1 \sin\theta, 0\right) \\ P_2' = \left(\frac{M\gamma}{2}, p_1 \cos\theta, -p_1 \sin\theta, 0\right) \end{cases}$$

$$P_1^{\prime 2} = -\frac{M^2\gamma^2}{4} + p_1^2 \cos^2\theta + p_1^2 \sin^2\theta = -m^2 \Rightarrow$$

$$\Rightarrow p_1^2 = \frac{M^2\gamma^2}{4} - m^2 \Rightarrow \boxed{p_1 = \sqrt{\frac{M^2\gamma^2}{4} - m^2}}$$

$$p = M\gamma v = 2 p_1 \cos\theta = \cos\theta \, 2\sqrt{\frac{M^2\gamma^2}{4} - m^2} = \cos\theta \sqrt{M^2\gamma^2 - 4m^2}$$

$$\Rightarrow \boxed{\cos\theta = \frac{M\gamma v}{\sqrt{M^2\gamma^2 - 4m^2}} = \frac{M\gamma v}{2\sqrt{\frac{M^2\gamma^2}{4} - m^2}}}$$

$$\Rightarrow p_1 \cos\theta = \cancel{\sqrt{\frac{M^2\gamma^2}{4} - m^2}} \frac{M\gamma v}{2\cancel{\sqrt{\quad}}} = \frac{M\gamma v}{2} \checkmark$$

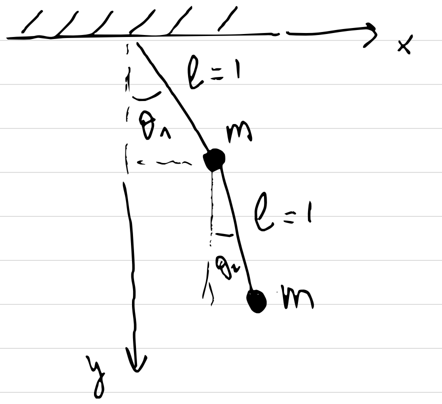
$$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{M^2\gamma^2 v^2}{\cancel{4}\frac{M^2\gamma^2}{\cancel{4}} - 4m^2}} =$$

$$= \sqrt{\frac{M^2\gamma^2 - 4m^2 - M^2\gamma^2 v^2}{M^2\gamma^2 - 4m^2}} = \sqrt{\frac{M^2\gamma^2(1-v^2) - 4m^2}{M^2\gamma^2 - 4m^2}} =$$

$$= \sqrt{\frac{M^2 - 4m^2}{M^2\gamma^2 - 4m^2}} ;$$

$$\text{tg}\,\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sqrt{\frac{M^2 - 4m^2}{M^2\gamma^2 - 4m^2}}}{\frac{M\gamma v}{\sqrt{M^2\gamma^2 - 4m^2}}} = \frac{\sqrt{M^2 - 4m^2}}{M\gamma v} \checkmark$$

E3



$$x_1 = \sin\theta_1 , \quad y_1 = \cos\theta_1$$

$$x_2 = x_1 + \sin\theta_2 , \quad y_2 = y_1 + \cos\theta_2$$

$$= \sin\theta_1 + \sin\theta_2 ; \qquad = \cos\theta_1 + \cos\theta_2$$

a)

$$\left.\begin{array}{l} \dot x_1 = \cos\theta_1 \, \dot\theta_1 \; ; \quad \dot x_1^2 = \cos^2\theta_1 \, \dot\theta_1^2 \\ \dot y_1 = -\sin\theta_1 \, \dot\theta_1 \; ; \quad \dot y_1^2 = \sin^2\theta_1 \, \dot\theta_1^2 \end{array}\right\} \dot x_1^2 + \dot y_1^2 = \dot\theta_1^2$$

$$\dot x_2 = \cos\theta_1 \dot\theta_1 + \cos\theta_2 \dot\theta_2 \; ; \quad \dot x_2^2 = \cos^2\theta_1 \dot\theta_1^2 + \cos^2\theta_2 \dot\theta_2^2 + 2\cos\theta_1 \cos\theta_2 \dot\theta_1 \dot\theta_2$$

$$\dot y_2 = -\sin\theta_1 \dot\theta_1 - \sin\theta_2 \dot\theta_2 \; ; \quad \dot y_2^2 = \sin^2\theta_1 \dot\theta_1^2 + \sin^2\theta_2 \dot\theta_2^2 + 2\sin\theta_1 \sin\theta_2 \dot\theta_1 \dot\theta_2$$

$$\Rightarrow \dot x_2^2 + \dot y_2^2 = \dot\theta_1^2 + \dot\theta_2^2 + 2\dot\theta_1\dot\theta_2 \left(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2\right)$$

$$= \dot\theta_1^2 + \dot\theta_2^2 + 2\dot\theta_1\dot\theta_2 \cos\left(\theta_1 - \theta_2\right)$$

Hint

$$\Rightarrow T = \frac{m}{2}\left[2\dot\theta_1^2 + \dot\theta_2^2 + 2\dot\theta_1\dot\theta_2\cos(\theta_1-\theta_2)\right]$$

$$V = -mg\,y_1 - mg\,(y_1 + y_2) = -mg\,(2\cos\theta_1 + \cos\theta_2)$$

$$\Rightarrow \boxed{\mathcal{L} = \frac{m}{2}\left[2\dot\theta_1^2 + \dot\theta_2^2 + 2\dot\theta_1\dot\theta_2\cos(\theta_1-\theta_2)\right] + mg\,(2\cos\theta_1 + \cos\theta_2)}$$

b) $\boxed{\theta_1}$

$$\frac{\partial\mathcal{L}}{\partial\theta_1} = -m\dot\theta_1\dot\theta_2 \sin(\theta_1-\theta_2) - 2mg\sin\theta_1$$

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot\theta_1}\right) = \frac{d}{dt}\left[2m\dot\theta_1 + m\dot\theta_2\cos(\theta_1-\theta_2)\right] =$$

$$= 2m\ddot\theta_1 + m\ddot\theta_2\cos(\theta_1-\theta_2) - m\dot\theta_2\sin(\theta_1-\theta_2)(\dot\theta_1-\dot\theta_2)$$

$$\Rightarrow 2\ddot\theta_1 + \ddot\theta_2\cos(\theta_1-\theta_2) - \dot\theta_2\sin(\theta_1-\theta_2)(\cancel{\dot\theta_1} - \dot\theta_2) =$$

$$= -\dot\theta_1\dot\theta_2\,\cancel{\sin(\theta_1-\theta_2)} - 2g\sin\theta_1 \Rightarrow$$

$$\Rightarrow \boxed{2\ddot\theta_1 + \ddot\theta_2\cos(\theta_1-\theta_2) + \dot\theta_2^2\sin(\theta_1-\theta_2) = -2g\sin\theta_1}$$

$\boxed{\theta_2}$

$$\frac{\partial\mathcal{L}}{\partial\theta_2} = +m\dot\theta_1\dot\theta_2\sin(\theta_1-\theta_2) - mg\sin\theta_2$$

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot\theta_2}\right) = \frac{d}{dt}\left[m\dot\theta_2 + m\dot\theta_1\cos(\theta_1-\theta_2)\right] =$$

$$= m\ddot\theta_2 + m\ddot\theta_1\cos(\theta_1-\theta_2) - m\dot\theta_1\sin(\theta_1-\theta_2)(\dot\theta_1-\dot\theta_2)$$

$$\Rightarrow \ddot\theta_2 + \ddot\theta_1\cos(\theta_1-\theta_2) - \dot\theta_1(\dot\theta_1 - \cancel{\dot\theta_2})\sin(\theta_1-\theta_2) =$$

$$= \dot\theta_1\dot\theta_2\,\cancel{\sin(\theta_1-\theta_2)} - g\sin\theta_2 \Rightarrow$$

$$\Rightarrow \boxed{\ddot\theta_2 + \ddot\theta_1\cos(\theta_1-\theta_2) - \dot\theta_1^2\sin(\theta_1-\theta_2) = -g\sin\theta_2}$$